



Elektrobit

MCAL Wrapper (McalExt) documentation

for CORTEXM S32K14X

product release 8.8.4



Elektrobit Automotive GmbH
Am Wolfsmantel 46
91058 Erlangen, Germany
Phone: +49 9131 7701 0
Fax: +49 9131 7701 6333
Email: info.automotive@elektrobit.com

Technical support

<https://www.elektrobit.com/support>

Legal disclaimer

Confidential information.

ALL RIGHTS RESERVED. No part of this publication may be copied in any form, by photocopy, microfilm, retrieval system, or by any other means now known or hereafter invented without the prior written permission of Elektrobit Automotive GmbH.

All brand names, trademarks, and registered trademarks are property of their rightful owners and are used only for description.

Copyright 2022, Elektrobit Automotive GmbH.

Table of Contents

1. Overview of MCAL Integration release notes	4
1.1. Location of MCAL documentation	4
2. Supported toolchain	5
2.1. Toolchain options	5
3. Scope of this release	6
3.1. Platforms Module	6
3.2. Third-party MCAL version	6
3.3. Third-party MCAL modules	6
3.4. Third-party MCAL patches by Elektrobit Automotive GmbH	7
3.4.1. Use of original or patched version in third-party MCAL modules	7
4. Overview of McalExt documentation	9
4.1. Background information	9
5. Using the McalExt module	10
5.1. Add McalExt and MCAL modules to project	10
5.2. EB build environment	11
5.3. User build environment	11
6. McalExt module description	12
6.1. Vendor delivery package	12
6.2. Connection with vendor delivery package	12
6.2.1. <code>plugin.xml</code> connection description	12
6.2.2. <code>Makefile</code> connection description	13
6.2.3. <code>anchors.xml</code> connection description	14
6.3. McalExt file description	15
6.4. MCAL version update	17

1. Overview of MCAL Integration release notes

Welcome to the MCAL release notes. These release notes are target-specific and derivative-specific.

[Chapter 2, “Supported toolchain”](#) provides information about the supported toolchain.

[Chapter 3, “Scope of this release”](#) provides specific information about the hardware-dependent third-party modules contained in this EB tresos AutoCore release:

- ▶ AUTOSAR version and revision of your hardware-dependent modules
- ▶ SWS version and revision
- ▶ Module version
- ▶ Supplier of your hardware-dependent modules

1.1. Location of MCAL documentation

Depending on the platform release that you purchased, some of the modules may be supplied to Elektrobit Automotive GmbH by third-parties. All MCAL modules are documented outside of these release notes. This documentation contains additional information about the third-party MCAL modules and the patches that were made by Elektrobit Automotive GmbH.

You can find the MCAL module documentation in the following locations:

- ▶ `$TRESOS_BASE/doc/5.0_MCAL_modules/MCAL_Wrapper_documentation.pdf`
- ▶ `$TRESOS_BASE/plugins/<McalExt plugin>/MCAL_Delivery/plugins/<Module name>/doc`
- ▶ In the online help of EB tresos Studio.

For information about the online help in EB tresos Studio, see the EB tresos Studio user documentation.

2. Supported toolchain

This release of EB tresos AutoCore supports MULTI_7.1.4_Comp_2017.1.4_ARM

2.1. Toolchain options

The toolchain options summarize under which conditions this release needs to be built. The release is tested using these toolchain options. If you change the toolchain options, consider this release *untested*.

Compiler	Options
ccarm.exe	<code>-cpu=cortexm4f, -fsingle, -ansi, -Osize, -dual_debug, -G, --no_exceptions, -Wundef, -Wimplicit-int, -Wshadow, -Wtrigraphs, -Wall, --prototype_errors, --incorrect_pragma_warnings, -noslashcomment, -nostartfile, --short_enum, --no_commons, -keeptempfiles, -list, -DGHS, -DDISABLE_MCAL_INTERMODULE_ASR_CHECK, -asm=-list, -Mn, -delete, -ignore_debug_references, -map, -keepmap, -preprocess_assembly_files, -c</code>
Assembler	Options
ccarm.exe	<code>-cpu=cortexm4f, -fsingle, -ansi, -Osize, -dual_debug, -G, --no_exceptions, -Wundef, -Wimplicit-int, -Wshadow, -Wtrigraphs, -Wall, --prototype_errors, --incorrect_pragma_warnings, -noslashcomment, -nostartfile, --short_enum, --no_commons, -keeptempfiles, -list, -DGHS, -DDISABLE_MCAL_INTERMODULE_ASR_CHECK, -asm=-list, -Mn, -delete, -ignore_debug_references, -map, -keepmap, -preprocess_assembly_files, -c</code>
Linker	Options
ccarm.exe	<code>-cpu=cortexm4f, -fsingle, -ansi, -Osize, -dual_debug, -G, --no_exceptions, -Wundef, -Wimplicit-int, -Wshadow, -Wtrigraphs, -Wall, --prototype_errors, --incorrect_pragma_warnings, -noslashcomment, -nostartfile, --short_enum, --no_commons, -keeptempfiles, -list, -DGHS, -DDISABLE_MCAL_INTERMODULE_ASR_CHECK, -asm=-list, -Mn, -delete, -ignore_debug_references, -map, -keepmap, -preprocess_assembly_files, -v</code>

3. Scope of this release

3.1. Platforms Module

This release of the `Platforms` module contains the mandatory and derivative-specific implementation part of the `Base` module.

This `Platforms` module shall be used only for S32K14X derivatives.

This module is tested only on hardware with the same sub-derivative as the third-party MCAL version. Other sub-derivatives are not tested.

3.2. Third-party MCAL version

This release contains the MCAL release S32K14X_MCAL4_2_RTM_HF2_1_0_2 from NXP.

This release of EB tresos AutoCore is tested only with sub-derivative s32k146_lqfp100.

3.3. Third-party MCAL modules

This release includes the hardware-dependent third-party MCAL modules listed in the table below.

Module name	AUTOSAR version and revision	SWS version and revision	Module version	Supplier
Adc	4.2.2	4.2.2	1.0.2	NXP
Can	4.2.2	4.2.2	1.0.2	NXP
Crcu	0.0.0	4.2.2	1.0.2	NXP
Dio	4.2.2	4.2.2	1.0.2	NXP
Eep	4.2.2	4.2.2	1.0.2	NXP
Eth	4.2.2	4.2.2	1.0.2	NXP
Fee	4.2.2	4.2.2	1.0.2	NXP
Fls	4.2.2	4.2.2	1.0.2	NXP
Gpt	4.2.2	4.2.2	1.0.2	NXP

Module name	AUTOSAR version and revision	SWS version and revision	Module version	Supplier
I2c	4.3.1	4.2.2	1.0.2	NXP
Icu	4.2.2	4.2.2	1.0.2	NXP
Lin	4.2.2	4.2.2	1.0.2	NXP
Mcem	4.2.2	4.2.2	1.0.2	NXP
Mcl	4.2.2	4.2.2	1.0.2	NXP
Mcu	4.2.2	4.2.2	1.0.2	NXP
Ocu	4.2.2	4.2.2	1.0.2	NXP
Port	4.2.2	4.2.2	1.0.2	NXP
Pwm	4.2.2	4.2.2	1.0.2	NXP
Spi	4.2.2	4.2.2	1.0.2	NXP
Wdg	4.2.2	4.2.2	1.0.2	NXP
Resource	0.0.0	4.2.2	1.0.2	NXP

Table 3.1. Third-party hardware-dependent modules

3.4. Third-party MCAL patches by Elektrobit Automotive GmbH

3.4.1. Use of original or patched version in third-party MCAL modules

The EB tresos Installer is the installation tool for the third-party MCAL modules. For more information on the EB tresos Installer, see [1.1_EB_tresos_installation_guide.pdf](#). In the EB tresos Installer, you can choose one of the following options:

- ▶ During installation it is possible to disable the update package. Afterwards, only original MCAL modules are part of your installation. For more information, see [Chapter 6, “McalExt module description”](#).
- ▶ If you also install the update package, all changes by Elektrobit Automotive GmbH that are described in [Section 6.3, “McalExt file description”](#) are part of your EB tresos AutoCore installation.

Additionally, with this update package you can switch between the original version and the patched version of each MCAL module.

The update package includes the `perform_MCAL_change.bat` batch file in the `McalExt` module. Use this batch file to switch from one version to another.

In the `perform_MCAL_change.bat`, you need to specify one of the following parameters:

- ▶ `EB_update`: Update the files in the module with content of Elektrobit Automotive GmbH.
- ▶ `origin`: Reset the files in the module with vendor content.

WARNING



Changes due to execution of the batch file affect all projects

The changes affect all projects that use the changed module. Therefore, execute the batch-file before generating code for your project.

4. Overview of McalExt documentation

This documentation is target-specific and derivative-specific.

This user guide describes the concepts and the configuration of the module:

► McalExt

4.1. Background information

McalExt is a wrapper module that connects the vendor MCAL delivery, EB tresos Studio, and the Elektrobit Automotive GmbH (EB) build environment. It allows you to use the vendor MCAL with EB tresos Studio with as few modifications as possible.

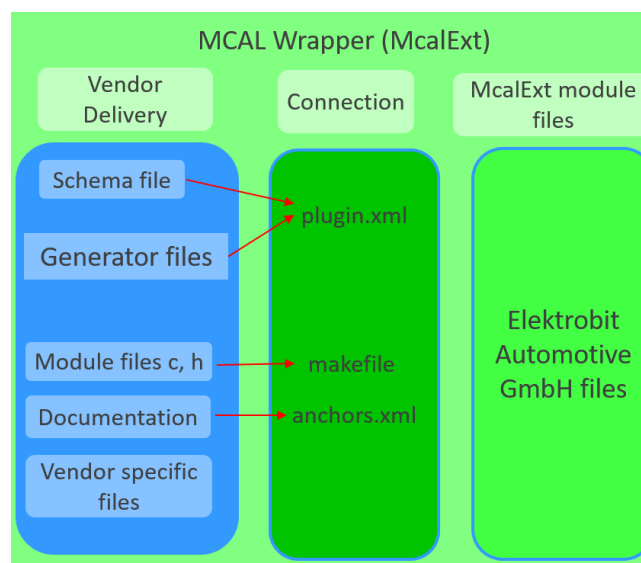


Figure 4.1. MCAL Wrapper (McalExt) overview

The MCAL Wrapper (McalExt) documentation also:

- provide information about the vendor delivery package, see [Section 6.1, “Vendor delivery package”](#)
- provide information about the connection with the vendor delivery package, see [Section 6.2, “Connection with vendor delivery package”](#)
- provide information about Elektrobit Automotive GmbH files, see [Section 6.3, “McalExt file description”](#)

5. Using the McalExt module

5.1. Add McalExt and MCAL modules to project

- Add the `McalExt` module to the project configuration.

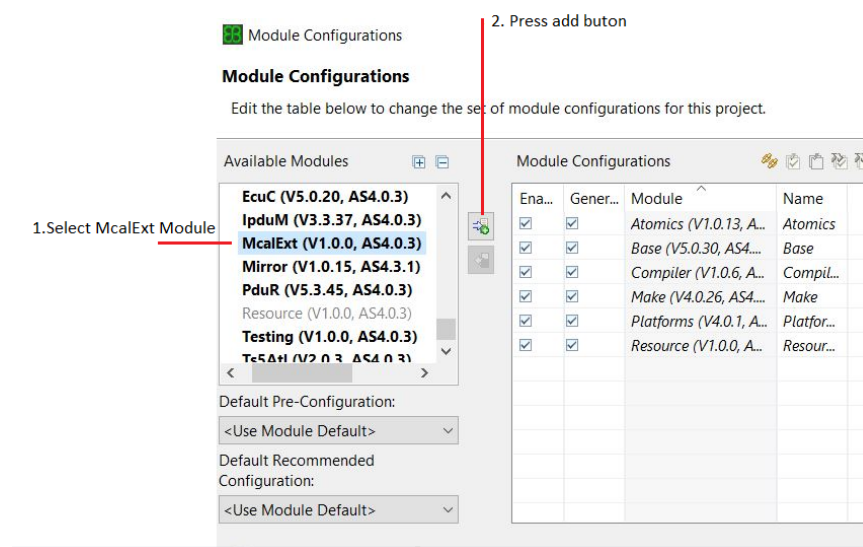


Figure 5.1. Add McalExt Module

- Add the needed MCAL modules to the project configuration

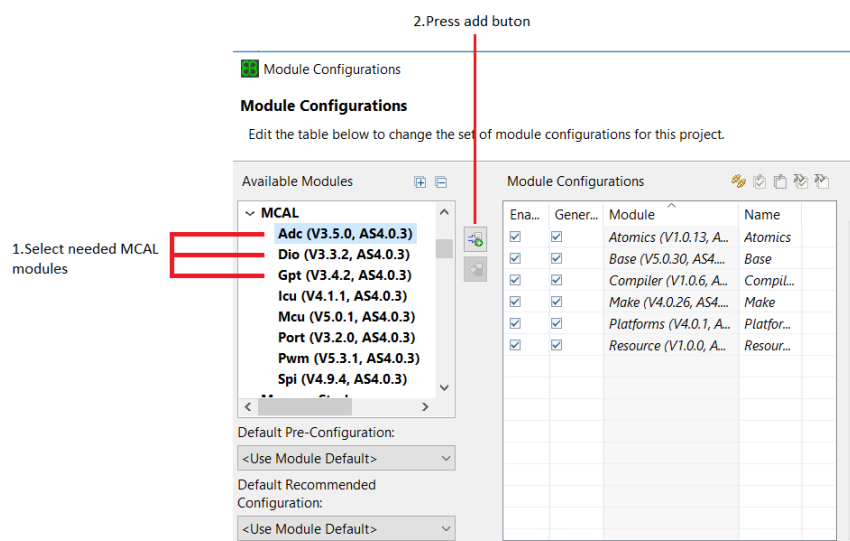


Figure 5.2. Add Mcal Module

5.2. EB build environment

In order to use the `McalExt` module in a project that is using the EB build environment, you must do the following:

- ▶ Add the `McalExt` module to the project configuration. see [Figure 5.1, “Add McalExt Module”](#)
- ▶ Add the needed MCAL modules to the project configuration. see [Figure 5.2, “Add Mcal Module”](#)

5.3. User build environment

When a different build environment (other than the EB build environment) is used in the project, you must do the following:

- ▶ Add the `McalExt` module to the project configuration. see [Figure 5.1, “Add McalExt Module”](#)
- ▶ Add the needed MCAL modules to the project configuration. see [Figure 5.2, “Add Mcal Module”](#)
- ▶ Add all the files that are configured in the module makefiles and all files configured in the `IncludePaths` and `FilesToBuild` from `McalExt` module configuration, see `IncludePaths` and `FilesToBuild` description in [Section 6.3, “McalExt file description”](#)

6. McalExt module description

6.1. Vendor delivery package

This represents the package that is delivered by different MCAL vendors (e.g. Infineon, Renesas, NXP, etc.) and contains different MCAL modules. For the MCAL modules and version that are integrated in this release please see [Section 3.3, “Third-party MCAL modules”](#).

6.2. Connection with vendor delivery package

The vendor delivery has a structure that cannot be used directly in the EB build environment. The `McalExt` module is introduced to make the connection between the EB build environment and the vendor delivery. The connection is made in the following files of the `McalExt` module:

`plugin.xml` connection, see [Section 6.2.1, “plugin.xml connection description”](#)

`makefile` connection, see [Section 6.2.2, “Makefile connection description”](#)

`anchors.xml` connection, see [Section 6.2.3, “anchors.xml connection description”](#)

6.2.1. plugin.xml connection description

- Schema file, e.g.:

```
<schema>
<manager class="dreisoft.tresos.autosar2.resourcehandling.AutosarSchemaManager"/>
<!-- Define the file(s) from which to load the schemas -->
<resource value="MCAL_Delivery/PathToSchemaFile/ModuleName.xdm" type="xdm"/>
</schema>
```

PathToSchemaFile - represents the path to where the module schema file is located in the MCAL vendor delivery.

ModuleName - represents the name of the schema file that should be used.

- Code Generator e.g.:

```
<!-- common template path parameters -->
```

```
<parameter name="templates"
mode="generate,verify" value="MCAL_Delivery/PathToGenerator"/>
```

PathToGenerator - represents the path where the generator is located in the vendor delivery.

- ▶ Ant code generator, e.g.:

```
<generator moduleId="ModuleId"
class="dreisoft.tresos.generator.ng.api.NGGenerator"
id="ModuleId_UniqueNGGeneratorId"
step="post"> <!-- run after code-generation -->
<parameter name="buildfile" value="MCAL_Delivery/PathToAntGenerator/AntGeneratorFile.xml"/>
</generator>
```

PathToAntGenerator - represents the path where the ant generator file is located in the vendor delivery.

AntGeneratorFile - represents the name of the ant generator file delivered by the vendor.

NOTE

This is applicable only if vendor provided the `AntGeneratorFile.xml` file.



6.2.2. Makefile connection description

For each MCAL module that is integrated in the `McalExt` a `make` folder exists that contains the makefiles for the respective MCAL module:

- ▶ `Module_defs.mak` file - registers the file(s) that are present in vendor delivery and the files that are generated for this module.

```
McalExt_GEN_FILES += $(McalExt_OUTPUT_PATH)\inc\ModuleName_Cfg.h
McalExt_GEN_FILES += $(McalExt_OUTPUT_PATH)\inc\ModuleName_PBcfg.h
McalExt_GEN_FILES += $(McalExt_OUTPUT_PATH)\src\ModuleName_PBcfg.c
CC_INCLUDE_PATH   += $(McalExt_CORE_PATH)\MCAL_Delivery\PathToHeaderFiles
```

ModuleName - represents the name of the header files that are generated.

PathToHeaderFiles - represents the path from the vendor delivery where the static header files are located.

- ▶ `Module_rules.mak` file - registers the specific module file(s) that are needed for compilation.

```
Module_src_FILES += $(McalExt_CORE_PATH)\MCAL_Delivery\PathToSourceFile\Module_Name.c
Module_src_FILES += $(McalExt_OUTPUT_PATH)\src\ModuleName_PBcfg.c
```

ModuleName - represents the name of the C files that are generated or are present in the vendor delivery.

PathToSourceFile - represents the path from the vendor delivery where the C static files are located.

All the MCAL modules makefiles will be included in `McalExt_defs.mak` and `McalExt_rules.mak` files only if the respective module is used in the EB tresos Studio project:

► **McalExt_defs.mak file**

```
ifeq ($(McalExt_Can_USED),true)
ifeq ($(Can_VARIANT),ModuleNameVariant)
include $(McalExt_CORE_PATH)\make\make_Can\Can_defs.mak
endif
endif
```

► **McalExt_rules.mak file**

```
ifeq ($(McalExt_Can_USED),true)
ifeq ($(Can_VARIANT),ModuleNameVariant)
LIBRARIES_TO_BUILD      += Can_src
include $(McalExt_CORE_PATH)\make\make_Can\Can_rules.mak
endif
endif
```

ModuleNameVariant - represents the MCAL module name that is used in the EB tresos Studio project.

6.2.3. anchors.xml connection description

► **anchors.xml file registers the MCAL documentation that is shown in the EB tresos Studio help window, e.g:**

```
<topic label="DocName" href="PathToDoc/DocName"/>
<topic label="DocName" href="PathToDoc/DocName"/>
<topic label="DocName" href="PathToDoc/DocName"/>
```

PathToDoc - represents the path where the MCAL documents are located.

DocName - represents the name of the MCAL documents.

6.3. McalExt file description

Some patches made by Elektrobit Automotive GmbH are due to missing or incomplete files in the MCAL vendor delivery. Additionally, some EB tresos Studio features are enabled. These patches are separated from the original installation files.

- ▶ The `McalExt` wrapper schema file `McalExt.xdm` located in `config` folder allows you to configure the following parameters that can be used in the project:
 - ▶ **PlatformModuleDefine** – you can configure defines that will be generated in `Platforms_Modules.h` file and that can be used in the project, e.g.: configure an `Mcu` configuration pointer that will be used in the `Mcu_Init()` function.

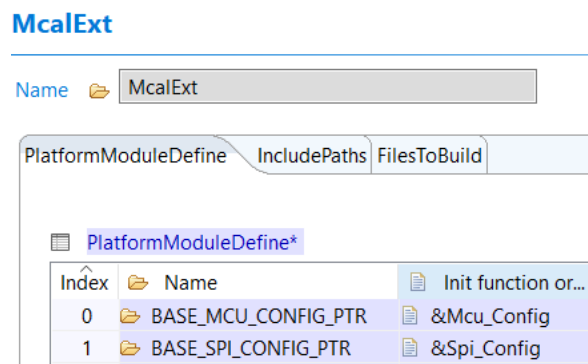


Figure 6.1. Platform Module Define parameter configuration

Generated file content example:

```
#define BASE_MCU_CONFIG_PTR          &Mcu_Config
#define BASE_SPI_CONFIG_PTR          &Spi_Config
```

Usage of defines (`EcuM_DriverInitListOne()`), e.g.:

```
/* *** Call service Init of module Mcu *** */
Mcu_Init(BASE_MCU_CONFIG_PTR);
```

- ▶ **IncludePaths** – allows you to configure different paths that need to be included by the build environment. This will be generated in the `McalExtWrapper.mak` file:

McalExt

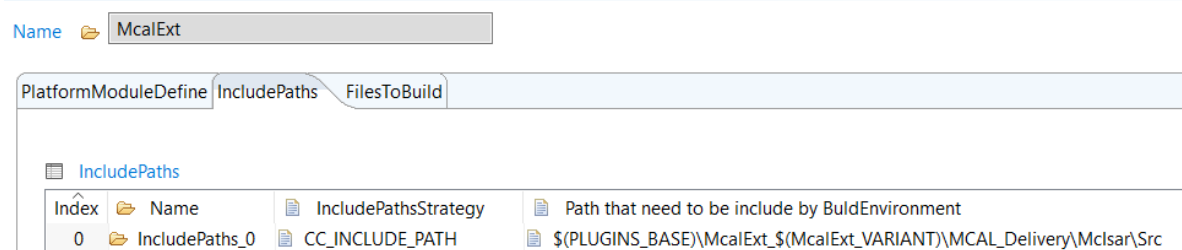


Figure 6.2. Path that needs to be included by BuildEnvironment

Generated content in `McalExtWrapper.mak` file:

```
CC_INCLUDE_PATH += $(PLUGINS_BASE)\McalExt_$(McalExt_VARIANT)\MCAL_Delivery\McIsar\Src
```

- **FilesToBuild** - allows you to configure different files that need to be compiled. This will be generated in the `McalExtWrapper.mak` file:

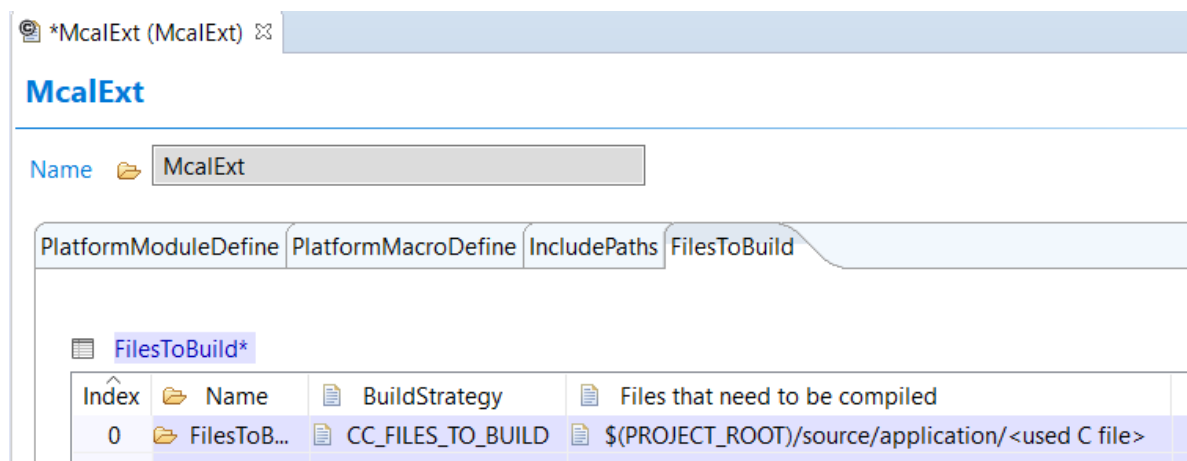


Figure 6.3. Files that need to be compiled

Generated content in `McalExtWrapper.mak` file:

```
CC_FILES_TO_BUILD += $(PROJECT_ROOT)\source\application\Eb_Intgr_BswM_UserCallouts.c
```

- Preconfiguration and recommended configuration – The `config_ext` folder can contain files for different MCAL modules, e.g:
 - Preconfiguration files that contain the parameter configuration value that should not be modified e.-g. `McuResetReasonConf`.

- ▶ Recommended configuration – the configuration that was validated by Elektrobit Automotive GmbH while performing IP3/QP2 (if this was ordered), e.g.: Mcu recommended configuration that contains the clock configuration and other Mcu-related parameter configuration. You can decide if the recommended configuration is used or not.
- ▶ `swcd` - includes the BSWMD files that are mandatory since AUTOSAR 4.0. Those files are used by BSW modules and EB tresos Studio wizards provided by Elektrobit Automotive GmbH.
 - ▶ Generation of exclusive areas in the EB tresos AutoCore Generic `Rte` module.
 - ▶ Mapping of the Main function in the EB tresos AutoCore Generic `Rte` module.
 - ▶ Generation of MemMap header file(s) in EB tresos AutoCore Generic `MemMap` module.
- ▶ `include` - contains header file(s) that are created/patched by Elektrobit Automotive GmbH.
- ▶ `src` - contains source file(s) that should be compiled, created by Elektrobit Automotive GmbH.
- ▶ `resources` - includes several XML-based service needs assistant or properties files that are provided by Elektrobit Automotive GmbH. These files support you to faster complete a valid configuration.
 - ▶ `Dem_Events.xml` – Dem event generation in the EB tresos AutoCore Generic `Dem` module.
 - ▶ EcuM initialization in the EB tresos AutoCore Generic `EcuM` module.
 - ▶ SchM Main function handling in the EB tresos AutoCore Generic `Rte` module.

6.4. MCAL version update

When the MCAL version is updated, you must verify the following:

- ▶ Check if all the used files are at the same location. If the vendor does not modify the MCAL installed structure, then the used path should be the same.
- ▶ During integration, Elektrobit Automotive GmbH applies patches on the vendor files due to some bugs or due to some incompatibility. These patches are located in the files with the `.EB_update` extension. When you install the new MCAL version, you must to verify if those patches need to be applied on the new MCAL version files.